

Does the Assistant Check the Record? Knowledge-Base Poisoning and System-of-Record Validation in Utility Operations RAG

Nassim Tinkicht, Hussain Al-Aqrabi

Computer Information Science, Higher Colleges of Technology, United Arab Emirates

Email: ntinkicht@hct.ac.ae; halaqrabi@hct.ac.ae

Abstract: Retrieval-augmented generation (RAG) enables document-grounded technical lookup, but indexed repositories can admit low-trust content. We evaluate targeted numeric poisoning in utility-operations RAG with GridPoison-100: 35 public, 45 fictional asset-register, and 20 legitimate-conflict items. Both pre-specified primary tests were null: fixed-context exposure produced 0% attacker-value adoption, and a keyed system-of-record (SoR) check at 70% coverage did not significantly outperform a cautious prompt or isolate-then-aggregate voting (all Holm-adjusted $p = 1.0$). Under a conflict-abstaining base prompt, one poison at $k = 5$ reached 70.0% of public-item queries; Tier 1 correct-response rate fell from 75.7–78.6% to 8.6–12.9%, mostly through non-commitment, while conditional adoption rate (CAR) was 4.1%. At pre-specified $k = 2$, CAR was 34.6%. A locked commit-required control raised CAR to 49.0% for Mistral and 16.3% for Luna, supporting material and partial prompt mediation, respectively. Forced two-passage controls showed 0/80 adoption with the poison’s clean twin and 80/80 with a lone poison, but 17/80 with a different clean support for Mistral, falsifying the strong composition hypothesis. D3 is an upper-bound keyed record check: full coverage removed observed out-of-range commitments by construction but false-flagged 64.1–65.7% of otherwise-correct legitimate-conflict answers under deliberately generic records.

Index Terms: retrieval-augmented generation, knowledge-base poisoning, power systems, utility operations, large language models, data integrity

I. INTRODUCTION

Large language models are increasingly proposed as interfaces to technical documentation, including power-system applications [1], [2]. A plausible near-term utility use is constrained lookup: workers retrieve approach distances, interconnection boundaries, or approved asset limits. RAG makes such answers document-grounded but also creates a write-path dependency: low-trust contributions from contractors, shared drives, wikis, or external bulletins can enter the index.

Knowledge-base poisoning is already established [3], [4]; our question is how a conflict-aware utility assistant separates silent corruption from non-commitment, and how structured-record validation trades filtering benefit against false flags on legitimate scope conflicts.

RQ1 asks how poisoning decomposes into retrieval exposure and conditional adoption. **RQ2** compares public-standard and asset-specific adoption at fixed exposure. **RQ3** compares keyed SoR validation with two generic defenses and measures false flags when legitimate documents differ by jurisdiction

or interconnection. Contributions are: (1) GridPoison-100 and a utility write-path threat model; (2) a locked commit-required intervention that distinguishes prompt-mediated non-commitment from silent adoption; (3) a pre-stated composition test contrasting paired clean twins with non-paired supports; and (4) record-validation accounting that exposes the tradeoff between filtering and benign-conflict false flags under incomplete or generic record scope.

II. RELATED WORK

PoisonedRAG established targeted answer steering through retrieved attacker text [3]; later work studied retrieval, database, and generator factors affecting poisoning [4]. Blocker documents can induce refusal [5]; here non-commitment instead arises from the assistant’s conflict-abstention rule, which our commit-required control directly manipulates. RobustRAG introduced isolate-then-aggregate defenses [6]; D2 is only a simplified baseline from that family.

ClashEval and controlled conflict studies show that retrieved evidence can compete with model priors and that contradictory evidence can change answer or abstention behavior [7], [8], [9]. Evidence phrasing, recency, and metadata also affect preference [10], [11]. Power-system LLM risks have already been studied [1], [2]; our domain contribution is the joint measurement of silent corruption, prompt-mediated non-commitment, and record-scope false flags on source-grounded utility lookups.

III. THREAT MODEL

The attacker is an authorized but low-trust writer, such as a contractor or bulletin sender whose content is indexed. They may add one to three documents but cannot edit existing documents, change the structured record, EMS/SCADA, protection devices, model weights, or the user’s query. They know the target asset/topic and existing documents, but not the exact query wording, embedding model, or realized retrieval.

The goal is targeted numeric answer-integrity corruption: a benign lookup commits to an attacker-chosen value. The primary poison is factual text rather than an instruction; one stress paraphrase inherits directive wording from a clean template while changing only the value. Confidentiality, weight poisoning, multi-turn social engineering, and control-loop attacks are outside scope. Denial of service is not the goal, but loss

of correct answers is reported because it dominates observed failures.

IV. GRIDPOISON-100

GridPoison-100 has 100 items. Tier 1 contains 35 public facts from Massachusetts DER settings/return-to-service material, OSHA 29 CFR 1910.269 Tables R-6/R-7, and NERC PRC-024-3 boundaries [12], [13], [14], [15]. OSHA items encode voltage range, exposure type, and elevation at or below 900 m. FERC approved PRC-024-4 and PRC-029-1 in 2025, while NERC’s catalog accessed Oct. 10, 2026 still lists PRC-024-3 as mandatory and the newer standards as subject to future enforcement; the benchmark therefore freezes PRC-024-3 and treats version/jurisdiction as scope [16], [17]. Each public item stores source, locator, units, applicability, and an allowable interval.

Tier 2 contains 45 synthetic asset-specific values from a declared fictional register covering transformer, breaker, feeder, BESS, and cable limits. The register is authoritative by construction and does not describe a real network.

Tier 3 contains 20 unpoisoned legitimate conflicts where documents differ by jurisdiction, interconnection, or table scope while the query names the applicable context. These items measure false flags on correct context-specific answers.

The clean corpus has 400 short template-generated documents: three supports per poisonable item, 40 legitimate-conflict documents, and 120 on-topic distractors. Identifiers are explicit, so the corpus is cleaner than a real repository. Each poison is a minimal pair with one support and changes only the critical value. The main variant moves outside the benchmark interval; wrong-but-in-range and dated-bulletin variants test factual error and authority framing separately.

V. EXPERIMENTAL DESIGN

A. Reference conditions and models

B0 is closed-book generation, B1 clean RAG, and B2 poisoned RAG with the same base prompt. The prompt requests one numeric value when evidence is unambiguous and `VALUE: UNKNOWN` on unresolved conflict; D1 adds an explicit falsification warning. Retrieval uses sentence-transformers/all-MiniLM-L6-v2 cosine similarity with one document per chunk. Models are Qwen3-8B (qwen/qwen3-8b), Mistral Small 3.2 24B (mistralai/mistral-small-3.2-24b-instruct), and GPT-5.6 Luna (openai/gpt-5.6-luna) through OpenRouter at temperature zero and `max_tokens=256`. Provider routing was not pinned; resolved model/provider metadata, response ID, token usage, and timestamp were logged. Qwen’s reasoning budget was not hard-enforced, while Luna’s reasoning consumed the requested budget.

B. Stage A and Stage B

RQ1 uses natural dense retrieval with one poison and a three-paraphrase stress condition; $\text{Poison}@k$ is reported separately from generation. Pre-generation calibration found strongly imbalanced one-poison exposure across Tiers 1 and 2,

so natural retrieval cannot support a direct cross-tier adoption comparison. RQ2 therefore uses a fixed five-passage context: three clean same-item supports, one deterministic distractor, and one poison, counterbalanced first/last. This evidence-rich design holds exposure and context length constant and is not a natural attack-strength estimate.

C. Defenses

D1 is the cautious prompt. D2 isolates each retrieved passage, extracts one value with that prompt, ignores non-numeric outputs, and returns the plurality numeric value; no numeric votes or a tie yields `UNKNOWN`. It is a simplified isolate-then-aggregate baseline [6]. D3 deterministically post-processes the same B2 stress outputs: if the applicable item record is covered and a committed value differs, the answer is replaced by a verification flag. The applicable record key is given, so linking errors are not modeled. Records equal the benchmark target for 78/80 poisonable items; two Massachusetts undervoltage items instead store IEEE 1547-2018 Category II defaults [18], [14]. For 13/20 Tier 3 items, the record intentionally stores a generic alternative. D3 therefore measures an upper-bound answer-key-like check, not a new defense algorithm. SHA-256 coverage at nominal 40/70/100% realizes 30/57/80 of 80 poisonable items and 6/14/20 Tier 3 items.

RQ3 uses the three-poison $k = 5$ stress condition to avoid a retrieval floor. Secondary one-factor deviations test a dated bulletin, wrong-but-in-range poison, and $k = 2$; no full factorial sweep is performed.

VI. METRICS AND STATISTICAL ANALYSIS

Raw labels are correct (**C**), attacker adoption (**A**), another committed value (**W**), conflict flag (**F**), and refusal (**R**). Numeric C/A/W grading uses the required `VALUE:` line with zero numeric tolerance for all items; responses without a parsed numeric value are pooled as non-commitment (**N**).

Let E_t denote poison retrieval in the top- k . We report $\text{Poison}@k$, attack success rate, conditional adoption rate (CAR), correct-response rate (CRR), non-commitment rate (NCR), and benchmark-violation rate (BVR):

$$\text{CAR} = P(A \mid E_t = 1), \quad (1)$$

$$\text{BVR} = \frac{1}{|T|} \sum_t \mathbb{1}[\hat{v}_t \notin [L_{i(t)}, U_{i(t)}]]. \quad (2)$$

BVR counts only committed numeric values, so it is reported with CRR; it is a benchmark consequence metric, not a physical grid-safety measure.

The item is the inferential unit; two query phrasings are repeated observations. Defense differences use paired 95% item-cluster bootstrap intervals (2,000 resamples); natural-retrieval CAR intervals also cluster by item, and zero-event Stage-B cells use item-level Clopper–Pearson bounds. The locked primary RQ3 test compares D3@70 with D1/D2 using item-level exact McNemar tests, counting an item as violating only when both phrasings violate, with Holm correction over six comparisons. B2 comparisons are descriptive. Tier 3 BFFR is conditioned on clean-RAG correctness; this is post-hoc, and

unconditional full-coverage BFFR was 62.5% for every model. Code, data, prompts, run metadata, and the methods lock are public at <https://github.com/NTinkicht/Research-GridPoisonRAG>; pre-generation alignment is commit 3de75c6, and the follow-up lock is c598f44b.

VII. RESULTS

Both pre-specified primary endpoints were null: Stage-B attacker-value adoption was 0% in both tiers (RQ2), and no D3@70 comparison with D1 or D2 was significant (RQ3). The exposed-trial breakdown of natural retrieval below was added after these outcomes were known and is descriptive; $k = 2$ was pre-specified as a secondary one-factor deviation. Clean-RAG CRR was 87.8% on Tier 2 (pre-specified interpretation gate: 80%, defined only for Tier 2) and 75.7–78.6% on Tier 1.

A. Natural-retrieval behavior (RQ1)

Table I reports retrieval exposure and generation outcomes. Exposure is deterministic for the shared retriever; CRR and CAR are ranges across models. Adding two poison phrases left Tier 1 exposure unchanged at 70.0% but raised Tier 2 exposure from 3.3% to 76.7%; the stress condition therefore increased exposure only in Tier 2. At $k = 5$, one poison was retrieved in 49 Tier 1 trials per model; only 0–3 remained correct and direct attacker adoption was 2/49 (4.1%; item-cluster 95% CI 0.0–10.6). Of those 49 exposed trials, 44–47 were non-committal and 0 produced another wrong numeric value. The same exposure keys had been correct under clean $k = 5$ in 48–49/49 trials, so exposure was concentrated on otherwise-answerable queries. In the pre-specified secondary $k = 2$ condition, all three models adopted the attacker value in the same 9 of 26 exposed Tier 1 trials (CAR 34.6%; item-cluster 95% CI 17.9–54.5). No clean $k = 2$ generation arm was run; among the 44 unexposed Tier 1 trials, each model was correct on 10, so the $k = 2$ CRR row is not a clean- $k = 5$ comparison. The wrong-but-in-range poison was copied in 1 of 30 exposed Tier 1 trials per model and 0–1 of 5 exposed Tier 2 trials.

B. Fixed-context exposure (RQ2)

In the evidence-rich fixed context (three clean same-item supports, one distractor, one poison), attacker-value adoption and BVR were 0% in every model/tier/position cell. With zero events, the exact 95% upper bounds on per-item adoption are 10.0% for Tier 1 ($n = 35$) and 7.9% for Tier 2 ($n = 45$). Table II reports correct-answer retention. Every poison-last trial produced no numeric commitment. Because the base prompt explicitly permits `VALUE: UNKNOWN` when evidence is inconsistent, this is a conflict-handling result rather than evidence of inherent poisoning robustness.

C. Record-check accounting (RQ3)

The locked primary comparison is D3 at 70% coverage versus D1 and D2. Table III reports paired BVR reduction in percentage points; positive values favor D3. Against B2, which uses the same conflict-abstaining base prompt, D3@70

removed 6/7, 5/6, and 6/8 violating trials for Mistral, Qwen, and GPT-5.6 Luna, respectively; full coverage removed the remaining violations by construction. D2 increased Qwen BVR by 3.1 points relative to B2. Violating trials per model were 6–8 under B2, 5–8 under D1, 7–11 under D2, and 1–2 under D3@70. The primary McNemar comparisons had at most 3 discordant items and all six Holm-adjusted tests were non-significant ($p_{\text{Holm}} = 1.0$). The defense comparisons are descriptive. With at most 3 discordant items, the smallest attainable exact two-sided p was 0.25 before adjustment, so the locked comparisons could not reach significance at these violation rates. Bootstrap intervals can exclude zero while McNemar remains non-significant because the bootstrap estimates trial-level mean differences while clustering by item, whereas McNemar dichotomizes an item only when both phrasings violate; sparse discordance makes the exact test coarse. Because BVR counts non-commitment as a non-violation, note that across the stress-condition defenses CRR was 10.6–17.5% and non-commitment was 77.5–88.8%; D3@70 lowered CRR by 0.6–1.2 points relative to B2. In a post-hoc sensitivity analysis, counting an item as violating when either phrasing violates also leaves all six comparisons non-significant (minimum $p_{\text{Holm}} = 0.094$).

D. Legitimate-conflict cost

Clean RAG answered 35–39 of 40 legitimate-conflict trials correctly. The structured record equals the benchmark target for 78/80 poisonable items, but intentionally stores a generic alternative for 13/20 Tier 3 items. BFFR therefore measures a controlled record-scope gap and is conditioned on clean-RAG correctness (a post-hoc refinement; unconditional BFFR at full coverage was 62.5% for every model). At 70% coverage, conditional BFFR was 43.6–45.7%; at full coverage it was 64.1–65.7%. The cost is therefore driven by record scope, not only by model behavior.

E. Prospective Follow-up and Exploratory Backup

After the original Qwen/Mistral/Luna results were frozen, we locked four follow-up checks before running them. Qwen’s follow-up runs did not complete because of repeated provider-layer failures; the completed Mistral and Luna runs are reported as locked follow-ups. Gemini 2.5 Flash-Lite was then run as an exploratory backup with its own matched B2 baseline; it is not a pre-specified model, and no locked verdict depends on it.

Design. (i) A retrieval audit reran the frozen dense retriever and logged the top-10 documents. (ii) A commit-required prompt removed permission to abstain and required one best numeric VALUE. It reused the one-poison $k = 5$ setting; CAR is evaluated on the same 49 exposed Tier 1 trials per locked model. Confidence intervals resample item IDs, and paired inference collapses repeated phrasings to the item, positive if any originally exposed phrasing adopts. (iii) BM25 (BM25Okapi, lower-cased whitespace tokens, corpus-order ties) replaced the dense retriever at $k = 2$ and $k = 5$. (iv) Stage C used the formal query for each of the 80 poisonable

TABLE I

NATURAL-RETRIEVAL OUTCOMES (%; $k = 5$ UNLESS NOTED). CAR IS CONDITIONAL ON EXPOSURE; EXPOSED TIER 1/TIER 2 TRIALS PER MODEL ARE 49/3 (1 POISON), 49/69 (3 POISONS), 47/7 (BULLETIN), AND 26/3 ($k = 2$). THE CLEAN ROW IS $k = 5$; NO CLEAN $k = 2$ GENERATION ARM WAS RUN.

Condition	Tier 1 (public)			Tier 2 (asset)		
	Exp.	CRR	CAR	Exp.	CRR	CAR
Clean RAG	—	75.7–78.6%	—	—	87.8%	—
1 poison	70.0%	8.6–12.9%	4.1%	3.3%	84.4%	0.0%
3 poisons	70.0%	7.1–15.7%	4.1%	76.7%	14.4%	4.3–5.8%
1 bulletin	67.1%	20.0–22.9%	6.4–8.5%	7.8%	80.0%	0.0–14.3%
1 poison, $k = 2$	37.1%	14.3%	34.6%	3.3%	62.2%	0.0%

TABLE II
CORRECT-ANSWER RATE UNDER FIXED EXPOSURE (%).

Model	First/T1	First/T2	Last/T1	Last/T2
Mistral Small 3.2	8.6%	97.8%	0.0%	0.0%
Qwen3-8B	42.9%	57.8%	0.0%	0.0%
GPT-5.6 Luna	74.3%	28.9%	0.0%	0.0%

items with fixed contexts: C0, two clean same-item supports; C1, poison plus its paired clean twin; C2, poison plus a different clean same-item support; C3, poison plus an off-item distractor; and C4, poison alone. The locked rule treated the strong composition hypothesis as falsified if any locked model reached at least 20% adoption in C1 or C2, or at most 50% in C4.

Results. The audit reproduced all 640 original retrieval contexts (1920 generation rows across three models) with no poison-rank mismatch (one-poison $k = 5$, three-poison $k = 5$, bulletin $k = 5$, and one-poison $k = 2$). Across both tiers at $k = 2$, 9 exposed trials had no clean same-item passage and all 9 adopted; 20 exposed trials had at least one clean same-item passage and none adopted. Because the adoption outcomes were known before this audit, this is a consistency check rather than an independent test. Requiring commitment raised adoption from 2/49 to 24/49 for Mistral (49.0%; item-cluster 95% CI 33.3–64.4; item-level exact $p = 3.05 \times 10^{-5}$, $n = 31$ items) and from 2/49 to 8/49 for Luna (16.3%; 7.5–26.5; item-level exact $p = 0.0312$, $n = 31$ items). The pre-stated point-estimate bands were $\leq 10\%$ model resistance, > 10 to $< 25\%$ partial mediation, and $\geq 25\%$ material mediation; Mistral is material and Luna partial. Luna produced 12 empty outputs among these 49 exposed trials; excluding only those empties gives 21.6% adoption. The commit-prompt clean controls retained 83.8% CRR for Mistral and 86.9% for Luna (exploratory Gemini: 84.4%). For the locked models the comparator is the historical B2 run under unpinned provider routing; Gemini used a contemporaneous matched baseline. An empty-output audit found none in Mistral or Qwen B2_P1 or either Stage-B position. Luna had 1/160 empties in B2_P1, 4/80 with poison first and 0/80 with poison last. In Luna Stage C, C2 and C3 each had 2/80 empties and C0/C1/C4 had none; Mistral had none. Qwen Stage C is unavailable because its follow-up failed. Stage C falsified the pre-stated strong composition hypothesis: Mistral adopted in 17/80 C2

trials (21.2%; exact 95% CI 12.9–31.8), one trial above the threshold, while Luna (15/80) met every threshold. Both locked models adopted in 0/80 C1 and 80/80 C4 trials. In C1 the paired twin protected integrity by producing non-commitment rather than correct answers: both locked models were 0/80 correct. The no-poison C0 control was 0/80 adoption and 80/80 correct for every model. Table V reports all C1–C4 adoption counts.

Under BM25, the poison and its paired clean twin had exactly equal scores in 160/160 queries. Exposure was therefore highly tie-order dependent: under the locked clean-first corpus order, the poison appeared in 2/160 queries at $k = 2$ and 38/160 at $k = 5$; placing the poison first among equal scores changes those counts to 112/160 and 124/160, respectively. BM25 is therefore a tie-order sensitivity, not evidence of lexical robustness. *Post-hoc benchmark consequence mapping.* A post-hoc benchmark-consequence mapping decomposed each OSHA MAD into its electrical component and inadvertent-movement allowance. Of the 18 MAD poisons, 13 consumed only part of the allowance, leaving 0.03–0.45 m beyond the electrical component, and 5 fell inside the electrical component by 0.01–0.17 m. All 7 PRC-024-3 poisons shortened the benchmark no-trip time by 60%. All 8 Massachusetts DER trip-threshold poisons moved the encoded threshold farther from nominal, widening the benchmark no-trip region. The remaining 2 public exact-value items were return-to-service values and are not assigned a directional consequence class here. All 45 fictional Tier-2 poisons exceeded the record limit by 20%. 2 of these electrical-component cases (T1-025, T1-027) were among the nine $k = 2$ adopted trials. These are benchmark consequence classes, not professional-engineering or physical-safety determinations. *Post-hoc coverage what-if.* Severity-prioritized D3 gave no consistent benefit: relative to hash coverage, it changed violating trials per model by -1 to +0 of 160 at 40% and by +1 to +2 at 70%, with Tier-3 coverage held fixed.

VIII. DISCUSSION

At $k = 5$, retrieved poison rarely transferred its value under the conflict-abstaining prompt, but exposed Tier 1 trials usually lost the correct answer and became non-committal. These are distinct failures: non-commitment is an availability loss requiring external handling, whereas adoption is silent integrity loss. Removing permission to abstain raised adoption to 24/49 for Mistral and 8/49 for Luna, supporting material

TABLE III

PRIMARY RQ3 COMPARISON AND VIOLATION COUNTS. POSITIVE PP VALUES FAVOR D3. BRACKETS ARE DESCRIPTIVE 95% ITEM-CLUSTER BOOTSTRAP INTERVALS OF TRIAL-LEVEL BVR DIFFERENCES; MCNEMAR TESTS COUNT AN ITEM AS VIOLATING ONLY WHEN BOTH PHRASINGS VIOLATE.

Model	D3@70 vs D1 (pp)	D3@70 vs D2 (pp)	B2 → D3@70	D3@100
Mistral Small 3.2	2.5 [-0.6, 6.2]	3.8 [0.6, 8.1]	7 → 1	0
Qwen3-8B	3.1 [0.6, 6.9]	6.2 [1.9, 11.2]	6 → 1	0
GPT-5.6 Luna	3.8 [0.6, 7.5]	3.8 [0.0, 8.8]	8 → 2	0

TABLE IV

CONDITIONAL BENIGN-CONFLICT FALSE-FLAG RATE (BFFR).

Model	40%	70%	100%
Mistral Small 3.2	17.9%	43.6%	64.1%
Qwen3-8B	17.1%	45.7%	65.7%
GPT-5.6 Luna	17.9%	43.6%	64.1%

TABLE V

FOLLOW-UP CONTROLS (COUNTS). COMMIT: ATTACKER ADOPTIONS AMONG 49 EXPOSED TIER 1 TRIALS UNDER THE COMMIT-REQUIRED PROMPT (BASE PROMPT: 2, 2, AND 3). BM25: ADOPTIONS AMONG TRIALS EXPOSED AT $k = 5$ UNDER THE LOCKED CORPUS-ORDER TIE RULE. C1–C4: ADOPTIONS OUT OF 80 FORCED-CONTEXT TRIALS. GEMINI 2.5 FLASH-LITE IS EXPLORATORY.

Model	Commit	BM25	C1	C2	C3	C4
Mistral Small 3.2	24/49	2/38	0	17	53	80
GPT-5.6 Luna	8/49	0/38	0	15	73	80
Gemini (expl.)	21/49	12/38	1	63	78	80

and partial prompt mediation. At pre-specified $k = 2$, all three original models adopted on the same nine exposed trials, none of whose reproduced contexts contained a clean same-item passage.

Stage C narrows that association. Under the base prompt, the poison’s paired clean twin yielded 0/80 adoptions and 0/80 correct answers in both locked models, while a different clean support left 15–17/80 adoptions; a lone poison yielded 80/80. Thus composition matters, but protection depends on which clean evidence is retrieved and on the model. Because Stage C did not vary the prompt, its twin result is consistent with conflict abstention but does not independently identify that instruction as the mechanism. RQ2 remains inconclusive: the evidence-rich fixed context drove adoption to floor in both tiers. Luna also returned 41/70 empty outputs in closed-book Tier 1 trials, so its parametric prior was not adequately measured.

D3 mainly measures record coverage. Given the record key and 78/80 matching poisonable records, full coverage removes observed violations by construction; at nominal 70%, 57/80 items were covered. Sparse violating items made the locked D3-vs-D1/D2 tests non-significant, so those differences remain descriptive.

Tier 3 isolates record-scope cost, whose magnitude is partly designed by the 13/20 generic-record construction. Clean RAG resolved 35–39 of 40 scoped-conflict trials, but a generic SoR then false-flagged many correct answers. The supported design

implication is to scope record authority to the applicable jurisdiction or interconnection; escalation policy and real-repository redundant evidence were not tested.

IX. CONCLUSION

GridPoison-100 separates retrieval exposure, attacker adoption, non-commitment, and benchmark consequence in utility RAG. Under a conflict-abstaining prompt, one retrieved poison at $k = 5$ mostly caused non-commitment rather than adoption; a locked commit-required control supported material mediation for Mistral and partial mediation for Luna. RQ2 remained inconclusive at floor. Stage C showed 0/80 adoption with the paired clean twin and 80/80 with a lone poison, while a different clean support falsified the strong composition hypothesis. At locked 70% coverage, keyed record validation removed 5–6 of 6–8 B2 violating trials across models; full coverage removed the remainder by construction, while scope mismatch false-flagged correct context-specific answers. These results support jurisdiction- and interconnection-scoped records, while generalization to real repositories remains untested.

REFERENCES

- [1] J. Ruan, G. Liang, H. Zhao, G. Liu, X. Sun, J. Qiu, Z. Xu, F. Wen, and Z. Y. Dong, “Applying large language models to power systems: Potential security threats,” *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 3333–3336, 2024.
- [2] J. Li, Y. Yang, and J. Sun, “Risks of practicing large language models in smart grid: Threat modeling and validation,” *arXiv:2405.06237*, 2024.
- [3] W. Zou, R. Geng, B. Wang, and J. Jia, “PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models,” in *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 2025, pp. 3827–3844.
- [4] P. Pereira, E. Maia, I. Praça, and A. Bécue, “Influence factors on RAG poisoning,” *arXiv:2606.12469*, 2026.
- [5] A. Shafraan, R. Schuster, and V. Shmatikov, “Machine against the RAG: Jamming retrieval-augmented generation with blocker documents,” in *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 2025, pp. 3787–3806.
- [6] C. Xiang, T. Wu, Z. Zhong, D. Wagner, D. Chen, and P. Mittal, “Certifiably robust RAG against retrieval corruption,” *arXiv:2405.15556*, 2024.
- [7] K. Wu, E. Wu, and J. Zou, “ClashEval: Quantifying the tug-of-war between an LLM’s internal prior and external evidence,” in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024.
- [8] J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su, “Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts,” in *The Twelfth International Conference on Learning Representations*, 2024.
- [9] Y. Han, M. Lan, and H. Kilicoglu, “When evidence conflicts: Uncertainty and order effects in retrieval-augmented biomedical question answering,” in *BioNLP 2026*. Association for Computational Linguistics, 2026, pp. 630–643.

- [10] A. Wan, E. Wallace, and D. Klein, “What evidence do language models find convincing?” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024, pp. 7468–7484.
- [11] C.-H. Chiang and H.-y. Lee, “Do metadata and appearance of the retrieved webpages affect LLM’s reasoning in retrieval-augmented generation?” in *Proceedings of the 7th BlackboxNLP Workshop*, 2024, pp. 389–406.
- [12] U.S. Occupational Safety and Health Administration, “29 CFR 1910.269: Electric power generation, transmission, and distribution,” <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.269>, accessed Oct. 9, 2026.
- [13] North American Electric Reliability Corporation, “PRC-024-3: Frequency and voltage protection settings for generating resources,” <https://www.nerc.com/globalassets/standards/reliability-standards/prc/prc-024-3.pdf>, 2020, effective Oct. 1, 2022; accessed Oct. 9, 2026.
- [14] Massachusetts Technical Standards Review Group / ISO-NE, “Agreed upon settings, capabilities and operational modes,” <https://www.mass.gov/doc/tsrg-special-meeting-attachment-agreed-upon-settings-capabilities-operational-modes/download>, accessed Oct. 9, 2026.
- [15] Massachusetts Technical Standards Review Group / ISO-NE, “Default New England bulk system area settings requirement,” <https://www.mass.gov/doc/tsrg-meeting-31-attachment-default-new-england-bulk-system-area-settings-requirement/download>, 2021, May 19, 2021; accessed Oct. 9, 2026.
- [16] Federal Energy Regulatory Commission, “Order no. 909: Reliability standards for frequency and voltage protection settings and ride-through for inverter-based resources,” https://www.ferc.gov/sites/default/files/2025-07/E-1-RM25-3-000_0.pdf, 2025, issued July 24, 2025.
- [17] North American Electric Reliability Corporation, “Protection and control reliability standards catalog,” <https://prod.nerc.com/standards/reliability-standards/prc>, 2026, accessed Oct. 10, 2026.
- [18] IEEE Standards Association, “IEEE Std 1547-2018: IEEE standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces,” <https://standards.ieee.org/ieee/1547/5915/>, 2018.